OCR | GCSE Science / Chemistry | Higher Tier (HT)

Life Cycle Assessment (LCA): Smartphone Evaluation

Topic C6 / B6: Environmental Chemistry, Resource Security & Sustainability

6 Marks
EXTENDED RESPONSE (QER)

Candidate Name: ..... Class / Set: ..... Date: .....

1 Comparative evaluation of smartphone life cycle stages

A life cycle assessment (LCA) is carried out to assess the environmental impacts and sustainability of a product throughout its entire lifespan from cradle to grave. The table below gives data on the relative energy consumed, greenhouse gases released, financial costs, and main environmental impacts for the four stages in the life cycle of a smartphone.

Life Cycle Stage & Key Processes	Energy Used (MJ)	CO_2 Released (kg CO_2e)	Estimated Cost (£ per device)	Key Environmental & Resource Impacts
Stage 1 **Raw Materials** Mining copper, gold, lithium; drilling crude oil for plastics; quarrying silica	**220** ~23% of total	**18** Mining & smelting	**£45** Precious metals	Open-cast mining causes severe habitat destruction, deforestation, soil erosion, toxic acid mine drainage, and rapid depletion of finite non-renewable mineral reserves.
Stage 2 **Manufacturing & Packaging** Semiconductor chip etching, screen assembly, battery fabrication, packaging	**560** **~58% (Highest)**	**46** **~57% (Highest)**	**£180** Cleanrooms & robotics	Immense electricity consumption for cleanroom silicon lithography; emission of potent fluorinated greenhouse gases; industrial wastewater containing toxic chemical solvents.
Stage 3 **Product Use (3-Year Lifespan)** Daily battery recharging via mains power grid (~1000 full recharge cycles)	**150** ~16% of total	**12** Grid electricity	**£25** Electricity bills	Recharging depends on fossil fuel combustion (gas/coal) at power stations releasing CO_2 into the atmosphere. Battery degradation reduces performance, driving early replacement.
Stage 4 **End-of-Life Disposal** Option A: Municipal landfill dump Option B: Dismantling & metal recycling	**10 / 35** Landfill vs Recycling	**4 / 2** Net 2 kg if recycled	**£5 / £15** Landfill vs Recycling	**Landfill:** Toxic heavy metals (lead, arsenic, lithium) leach into soil and aquifers; non-biodegradable plastics persist. **Recycling:** Recovers scarce precious metals; avoids toxic runoff; offsets virgin mining.
Total (Full Life Cycle)	**940 – 965 MJ**	**76 – 80 kg CO_2e**	**£255 – £265**	***Cumulative cradle-to-grave environmental and carbon footprint per smartphone***

Evaluate the costs and environmental impacts associated with each of the four stages in the life cycle of a smartphone.

In your answer you should:

- Use data from the table to compare the **energy consumed** and **CO_2 released** across each stage.
- Evaluate the **financial costs** and **environmental trade-offs** at each stage, including disposal options.
- Give a **justified conclusion** stating which stage has the greatest overall impact and suggest how this could be reduced.

✍ Continuous prose: compare data (energy & CO_2), weigh economic vs environmental trade-offs, and justify your conclusion. **[6 Marks]**

[Total: 6 marks]

MARK SCHEME | OCR GCSE Science | Higher Tier (HT)

Teacher Mark Scheme: 6-Mark Smartphone LCA Evaluation

Max: 6

Assessment Objectives: AO1 (Demonstrate knowledge) · AO2 (Apply data & analyze) · AO3 (Evaluate, conclude & mitigate)

Level (Marks)	Generic Description	Indicative Level Criteria
Level 3 (5–6)	A comprehensive, well-structured evaluative response assessing both **environmental impacts** and **costs** across **all four stages** of the smartphone life cycle. Accurately uses and compares quantitative data (energy in MJ and CO_2 in kg) from the table. Thoroughly weighs up environmental and economic trade-offs (including disposal/recycling). Formulates a reasoned, justified conclusion identifying the stage with the greatest impact (manufacturing) and suggests plausible reduction strategies.	• Evaluates all 4 stages (impacts & costs). • Compares numerical data for energy (MJ) & CO_2 (kg). • Evaluates end-of-life trade-offs (recycling vs landfill). • Justified conclusion + realistic reduction strategies. • Coherent scientific terminology and structured prose.
Level 2 (3–4)	A sound evaluation covering at least **two or three stages**, or outlines all four stages with less depth. Includes some quantitative comparison of energy or CO_2 data from the table. Discusses environmental impacts or costs, with some awareness of trade-offs (e.g. recycling benefits or mining destruction). May provide a conclusion, but with weak justification or lacking reduction strategies.	• Evaluates 2–3 stages, or superficial coverage of all 4. • Quotes/compares some table data (e.g. manufacturing has highest energy/CO_2). • Discusses impacts or costs with some trade-off analysis. • Basic conclusion given without full justification.
Level 1 (1–2)	Identifies some life cycle stages and quotes isolated data or makes simple, isolated descriptions of environmental impacts or costs. Little or no comparative evaluation, trade-off analysis, or justified conclusion.	• Names 1–2 stages or quotes isolated numbers from the table. • Simple recall statements without synthesis. • No evaluation, trade-offs, or justified conclusion.
0 marks	No creditworthy response or content completely irrelevant.	No relevant scientific content.

🗒️ INDICATIVE SCIENTIFIC CONTENT (EVALUATION OF THE 4 STAGES, COSTS, IMPACTS & DATA)

⛏️ Stage 1: Raw Materials (220 MJ · 18 kg CO_2 · £45)

Environmental Impacts: Open-cast mining for copper, gold, lithium, and rare earths causes habitat loss, deforestation, soil erosion, and acid mine drainage/toxic chemical runoff into river ecosystems. Drilling crude oil for polymers risks spills and depletes finite fossil reserves.
Cost & Energy: High upfront purchasing costs (£45) due to mineral scarcity and global supply chains; smelting and purification require 220 MJ (23% of total).

🏭 Stage 2: Manufacturing & Packaging (560 MJ · 46 kg CO_2 · £180)

Dominant Stage: Accounts for over 58% of energy and 57% of total greenhouse gas emissions (highest of all stages).
Environmental Impacts: Silicon wafer etching and cleanroom microchip fabrication demand massive electrical power; releases potent fluorinated gases and toxic solvent wastewater.
Cost: By far highest financial cost (£180, ~68% of total device cost) due to advanced micro-engineering, cleanrooms, precision robotics, and packaging.

📱 Stage 3: Product Use Over 3 Years (150 MJ · 12 kg CO_2 · £25)

Environmental Impacts: Charging phone daily draws electricity from the national grid; fossil fuels (coal/gas) burned in power stations release 12 kg CO_2 into the atmosphere, contributing to global warming. Battery capacity degrades over time.
Cost: Very cheap to consumer (~£8/year, £25 total over 3 years). This low financial running cost often blinds consumers to the hidden embodied environmental cost of manufacturing.

♻️ Stage 4: Disposal Trade-offs (10–35 MJ · 2–4 kg CO_2 · £5–£15)

Option A (Landfill - £5, 4 kg CO_2): Cheap upfront, but toxic heavy metals (lead, arsenic, lithium) leach into groundwater, polluting ecosystems; non-biodegradable plastics persist; scarce finite metals are permanently lost.
Option B (Recycling - £15, 35 MJ, net 2 kg CO_2): Requires collection/sorting energy and higher initial cost (£15), but recovers finite valuable metals, conserves virgin ores, prevents toxic leaching, and reduces future Stage 1 extraction impact.

🎯 Evaluative Conclusion & Mitigation Strategies (Essential for Level 3):

- **Justified Judgment:** Stage 2 (Manufacturing) has by far the most severe environmental and financial impact (560 MJ energy, 46 kg CO_2, £180 cost).
- **Mitigation 1 (Extend Lifespan):** Keeping phones for 5–6 years instead of 2–3 halves annual manufacturing impact.
- **Mitigation 2 (Modular Design):** Designing phones with easily replaceable batteries and screens (right to repair) prevents premature discarding.
- **Mitigation 3 (Renewable Energy):** Powering semiconductor factories and server centres with 100% renewable solar/wind energy.
- **Mitigation 4 (Closed-Loop Recycling):** Mandatory manufacturer buy-back schemes to replace virgin mined ores with recycled metals.

Examiner Guidance & Common Pitfalls:

- **Evaluate Command Word:** Higher Tier Level 3 requires weighing up advantages/disadvantages, costs, and environmental impacts across all stages, backed by quantitative data, plus a supported conclusion. Mere description of data caps marks at Level 2.
- **Quantitative Comparison:** Candidates must compare specific values (e.g. manufacturing emits nearly 4× more CO_2 than use: 46 kg vs 12 kg).
- **Economic vs Environmental Trade-offs:** High marks require recognizing that the cheapest disposal option (landfill at £5) causes the greatest long-term ecological damage, whereas recycling (£15) conserves scarce finite resources.
- **Transport:** Transport occurs across all 4 stages and is embedded in the data (do NOT accept transport as a 5th stage).